

TrES-2b

R-2469

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_120908 · created 2026-08-23T23:12:58+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456178.71689 +/- 0.00018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.148 +/- 0.044
Transit depth (Rp/Rs)^2	2.19 +/- 1.3 [%]
Orbital Inclination (inc)	86.65 +/- 2.87
Airmass coefficient 1 (a1)	0.034 +/- 0.023
Airmass coefficient 2 (a2)	2.44 +/- 0.33
Scatter in the residuals of the lightcurve fit is	84.81 %
Best Comparison Star	None
Optimal Aperture	3.0
Optimal Annulus	5.5
Transit Duration (day)	0.095 +/- 0.028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.148 ± 0.044 vs 0.1254 (deviation 0.32σ)
Duration fit vs published	0.095 d vs 0.0746 d (deviation 0.46σ)
Mid-time O–C	-0.1 min
Depth SNR	2.1
Residual scatter	84.81 %

Light curve

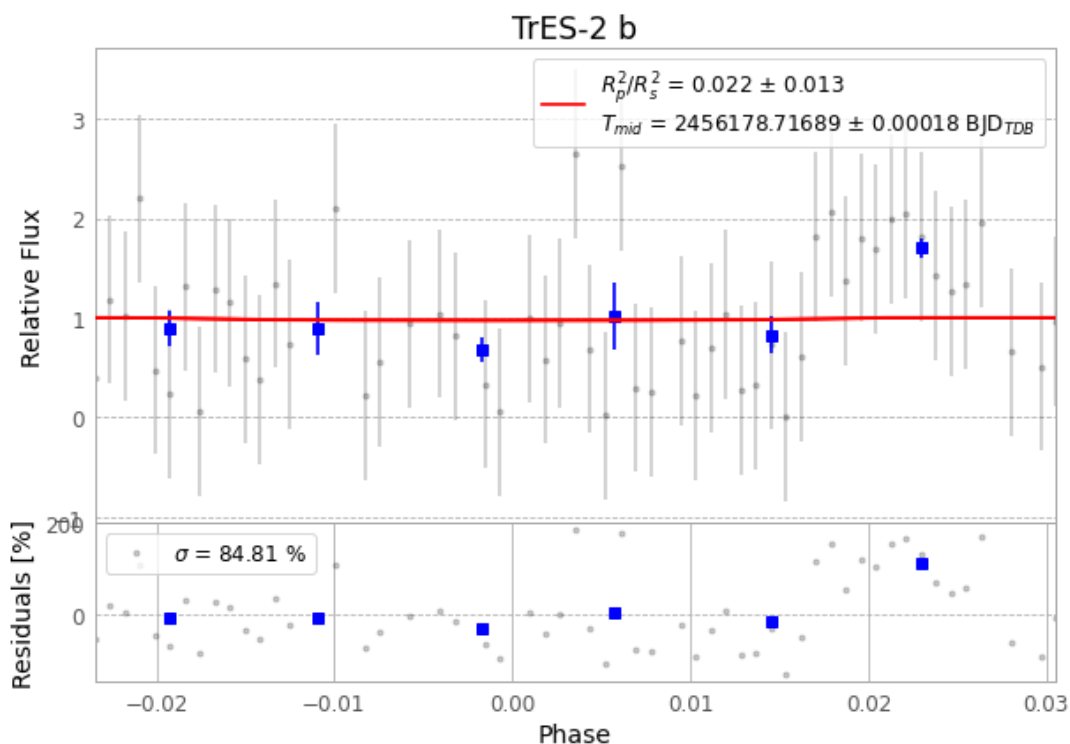


Figure 1 — FinalLightCurve_TrES-2 b_2012-09-07.png (SHA-256 ffa4b4a6037d8f16...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [636, 125], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f2aa00c345638051...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

65 FITS frames (43 MB), TRES-2120908034512.FITS → TRES-2120908065712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-120908.log (86e6806919e9...), FinalLightCurve_TrES-2 b_2012-09-07.png (ffa4b4a6037d...), FinalLightCurve_TrES-2 b_2012-09-07.csv (bc01bc6cdda0...)

Compute resources

Wall time 164.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-23 23:12Z.